



Student Number .....

# Barker College

## 2010 TRIAL HIGHER SCHOOL CERTIFICATE

# Chemistry

Staff Involved:

AM WEDNESDAY 4<sup>TH</sup> AUGUST

- RJP
- RZS \*
- KHW
- NJD

### SECTION I ANSWER SHEET

94 copies

Section I

Multiple Choice

Choose the best response and fill in the response oval completely

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|     |                         |                         |                         |                         |
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| 18. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 19. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| 20. | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |



Barker College

**2010**  
**TRIAL**  
**HIGHER SCHOOL**  
**CERTIFICATE**

# Chemistry

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- RJP
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- NJD

**AM WEDNESDAY 4<sup>TH</sup> AUGUST**

**94 copies**

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- Draw diagrams using pencil
- A Data Sheet and Periodic Table are provided at the back of this paper
- Write your Barker Student Number at the top of the Section I Answer Sheet, at the top of this page and at the top of **ALL** answer pages in Section II
- Show **ALL** working for calculations

**Total marks – 100**

### Section I

Pages 2 - 7

**20 marks**

- Multiple Choice
- Indicate all answers on the Answer Sheet provided
- Allow about 30 minutes for this section.

### Section II

Pages 8 - 28

**80 marks**

- Short Response
- Indicate all answers in the spaces provided on the paper
- Allow about 150 minutes for this section

## Section I

20 marks

### Attempt Questions 1-20

Allow about 30 minutes for this part

Use the multiple-choice answer sheet provided for Questions 1-20.

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1. Which of the following is classified as a biopolymer?
  - (A) PVC
  - (B) Polystyrene
  - (C) Glucose
  - (D) Cellulose
  
2. Which of the following species has the greatest ability to gain electrons from a reducing agent in solution?
  - (A) Zinc ions
  - (B) Silver ions
  - (C) Zinc
  - (D) Silver
  
3. Which of the following elements is classified as transuranic?
  - (A) Cerium
  - (B) Curium
  - (C) Caesium
  - (D) Cadmium
  
4. When glucose is fermented to form ethanol and carbon dioxide, the carbon dioxide produced can be bubbled into limewater,  $\text{Ca}(\text{OH})_2$  to form a white precipitate of calcium carbonate. What is the maximum mass of calcium carbonate which could be formed from 90 g of glucose?
  - (A) 100 g
  - (B) 50 g
  - (C) 200 g
  - (D) 90 g

## Section I continued

5. Some salts, when dissolved in water, produce acidic or alkaline solutions, while others produce neutral solutions. Which is the CORRECT statement?
- (A) When sodium chloride is dissolved in water, the pH increases slightly.
  - (B) When potassium ethanoate is dissolved in water, the concentration of hydroxide ions decreases.
  - (C) When sodium carbonate is dissolved in water, the pH decreases.
  - (D) When ammonium nitrate is dissolved in water, the concentration of hydronium ions increases.
6. An indication that an oxide is classified as basic is when it:
- (A) reacts with water to form a base or with a salt to form an acid.
  - (B) reacts with water to form an acid or with a base to form a salt.
  - (C) reacts with an acid to form a salt or with water to form hydroxide ions.
  - (D) turns litmus paper red.
7. Which of the following mixtures would form a buffer solution?
- (A) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOOH}$  and 1000 mL of  $0.1 \text{ mol L}^{-1} \text{CH}_3\text{COO}^- \text{Na}^+$
  - (B) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HNO}_3$  and 25 mL of  $1 \text{ mol L}^{-1} \text{NO}_3^-$
  - (C) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOOH}$  and 100 mL of  $0.1 \text{ mol L}^{-1} \text{CH}_3\text{COOH}$
  - (D) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOOH}$  and 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOO}^- \text{Na}^+$
8. The chemical added to bulk water supplies to make the solution alkaline is:
- (A) chlorine
  - (B) sodium fluoride
  - (C) lime
  - (D) iron (III) chloride

## Section I continued

9. Some measurements were made on various water samples and the results tabulated.

| Sample                   | A   | B   | C   | D   |
|--------------------------|-----|-----|-----|-----|
| DO (ppm)                 | 2.2 | 7.0 | 8.7 | 5.6 |
| TDS (ppm)                | 550 | 600 | 45  | 200 |
| Turbidity (NTU)          | 100 | 250 | 5   | 10  |
| pH                       | 7.5 | 3.5 | 6.5 | 7.5 |
| Coliforms (CFU / 100 mL) | 200 | 0   | 0   | 0   |

The sample most likely to have been drawn from a water supply polluted by run-off from a mining site is:

- (A) A  
(B) B  
(C) C  
(D) D
10. Which of the following is the conjugate acid of  $H_2PO_4^-$  ?

- (A)  $H_2PO_4$   
(B)  $H_3PO_4$   
(C)  $HPO_4^{2-}$   
(D)  $PO_4^{3-}$

11. Consider the four chloroalkanes

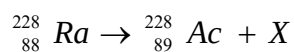
- 2, 4-dichloropentane
- 2, 4-dichlorohexane
- 2, 3-dichlorohexane
- 2, 2, 3, 4-tetrachloropentane

Which of the four chloroalkanes are isomers?

- (A) 1 and 2  
(B) 1 and 4  
(C) 2 and 3  
(D) 3 and 4

## Section I continued

12. Atomic absorption spectroscopy (AAS) would be a suitable method for determining the concentration of:
- (A) Phosphorus in detergents
  - (B) Ethanol in wine
  - (C) Mercury in fish
  - (D) Nitrogen in fertilisers
13. 0.1 mol/L citric acid is neutralised by a solution of 0.1 mol/L sodium hydroxide. Choose the best indicator for this titration.
- (A) Methyl orange
  - (B) Phenolphthalein
  - (C) Bromothymol blue
  - (D) Universal
14. Butanoic acid and ethyl butanoate have the same molecular formula,  $C_4H_8O_2$ . Which of the following is the best chemical test to distinguish between these two compounds?
- (A) Dilute sodium carbonate
  - (B) Bromine water
  - (C) Solubility in water
  - (D) Smell
15. The following equation represents the nuclear decay of radium - 88.



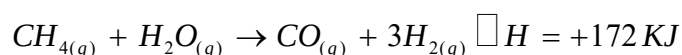
Identify the species marked X.

- (A) An alpha particle
- (B) A beta particle
- (C) A neutron
- (D) A proton

## Section I continued

16. The quantity of flocculent added to a water supply was less than the amount required to effectively treat that supply. What would be the effect on the water quality?
- (A) A high value of TDS (total dissolved solids)
  - (B) A pH between 5 and 6
  - (C) Turbid water
  - (D) A high bacterial count

17. One possible source of hydrogen for use of a fuel is through the “reforming” of natural gas. The key reaction is :



A catalyst is used for this reaction.

Which of the following is the most suitable set of conditions for this reaction?

- (A) Low pressure and low temperature
  - (B) Low pressure and high temperature
  - (C) High pressure and high temperature
  - (D) High pressure and low temperature
18. In an aqueous solution of the weak acid nitrous acid,  $\text{HNO}_2$ , which of the following species is present?
- (A)  $\text{HNO}_2$
  - (B)  $\text{H}_3\text{O}^+$
  - (C)  $\text{OH}^-$
  - (D)  $\text{NO}_2^-$
19. Which of the following reactions will occur as written?
- (A)  $\text{Cu}_{(s)} + \text{Zn}(\text{NO}_3)_{2(aq)} \rightarrow \text{Cu}(\text{NO}_3)_{2(aq)} + \text{Zn}_{(s)}$
  - (B)  $\text{Cu}_{(s)} + \text{Sn}(\text{NO}_3)_{2(aq)} \rightarrow \text{Cu}(\text{NO}_3)_{2(aq)} + \text{Sn}_{(s)}$
  - (C)  $2\text{Ag}_{(s)} + \text{Cu}(\text{NO}_3)_{2(aq)} \rightarrow 2\text{AgNO}_{3(aq)} + \text{Cu}_{(s)}$
  - (D)  $\text{Sn}_{(s)} + \text{Cu}(\text{NO}_3)_{2(aq)} \rightarrow \text{Sn}(\text{NO}_3)_{2(aq)} + \text{Cu}_{(s)}$

### **Section I continued**

- 20.** Why is the maximum yield of ethanol produced by the yeast - catalysed fermentation of 1 litre of glucose solution approximately 150ml?
- (A) The reaction is limited by the amount of water.
  - (B) There is too much heat generated if more ethanol is produced.
  - (C) Too much oxygen is generated if more ethanol is produced.
  - (D) The yeast can only tolerate an environment with a maximum 15% alcohol.

**End of Section I**

**Section II****Total marks 80****Attempt Question 21 - 36****Allow about 150 minutes for this section.****Question 21 (3 marks)****Marks**

A student prepared a sample of an indicator, using red cabbage. She tested the indicator with solutions of known pH and tabulated her observations.

| Colour | Red |   | violet |   | purple |   | blue |   | green |    | yellow |    |    |    |
|--------|-----|---|--------|---|--------|---|------|---|-------|----|--------|----|----|----|
| pH     | 1   | 2 | 3      | 4 | 5      | 6 | 7    | 8 | 9     | 10 | 11     | 12 | 13 | 14 |

- (a) State the colour of the indicator in a  $0.0005 \text{ mol L}^{-1}$  solution of sulfuric acid. Show your working.

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- (b) 1.00 mL of the  $0.0005 \text{ mol L}^{-1}$  solution of sulfuric acid was diluted to 1 L, using deionised water. What could be the colour of the diluted solution after addition of a few drops of red cabbage indicator? Show working.

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Page Total

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**Question 22** (5 marks)

**Marks**

Cobalt-60 (Co - 60) is a gamma emitter. It is produced by the neutron bombardment of Co-59.

- (a) Name an instrument that can be used to detect gamma radiation. 1

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- (b) Write a nuclear equation to show the formation of Co-60 from Co - 59. 1

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- (c) Evaluate the medical use of Co-60 or another identified medical radioisotope you have studied. 3

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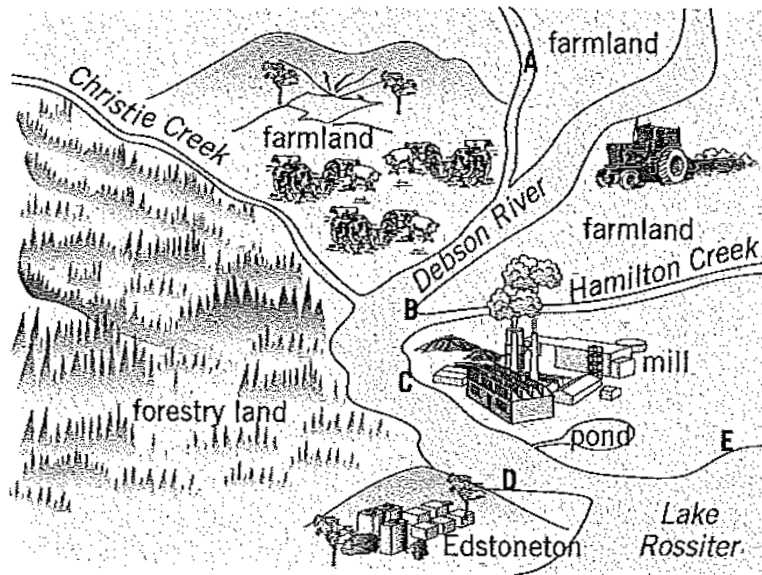
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**Question 23** (7 marks)

**Marks**

The map below shows the location of several sites which affect the water quality of the town, Edstoneton. Various water testing stations are labelled A - E.



- (i) The water temperature at testing station D is quite high. Account for this high temperature.

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- (ii) The water in the Debson River experiences algal blooms especially after heavy rain. Name the possible nutrient responsible for the algal bloom and state its likely source. 2

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- (iii) What effect could increased water temperature have on the level of dissolved oxygen in the river? Explain your answer. 2

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**Question 23 continues on page 11**

Page Total

**Question 23** (continued)

**Marks**

- (iv) Recently some of the forestry land along Christie Creek has been cleared for urban development. This has affected the water quality at Testing Station B particularly after heavy rain. What would an observer notice about a water sample at Testing Station B?

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- (v) Describe **ONE** water treatment method that could be used to sterilise the water at site A so that it would be suitable for drinking.

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Page Total

**Question 24** (5 marks)

Sodium hydrogen phosphate ( $\text{Na}_2\text{HPO}_4$ ) forms an amphiprotic species in water that is involved in the buffering of living cells.

- (a) Write an equation showing how  $\text{Na}_2\text{HPO}_4$  can act as an acid in water. 1

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- (b) Write an equation showing how  $\text{Na}_2\text{HPO}_4$  can act as a base in water. 1

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- (c) From one of your equations above clearly identify a conjugate acid-base pair. 1

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- (d) The behaviour of the above amphiprotic species ( $\text{HPO}_4^{2-}$ ) fits the definition of an acid / base theory you have studied. Identify the theory and explain how it fits that definition. 2

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**Question 25** (4 marks)

In class you analysed, by means of titration, a sample of fertiliser to determine the percentage of sulfate it contained. Below is a typical table of results.

| Titre | BaCl <sub>2</sub><br>Volume (mL) |
|-------|----------------------------------|
| 1     | 29.50                            |
| 2     | 29.45                            |
| 3     | 29.42                            |

9.32 g of the original fertiliser were dissolved in 250 mL of deionised water. 10 mL of this solution were pipetted into a conical flask. The burette contained 0.051 mol L<sup>-1</sup> barium chloride.

Calculate the percentage of sulfate in the fertiliser. Show all your working as a means of explaining the chemistry involved.

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**Question 26 (7 marks)**

**Marks**

The Haber process was developed in 1908 as a method of synthesising ammonia and was converted into a successful industrial process during the following decade.

- (a) Evaluate the significance of the Haber process at that time in world history. **2**

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- (b) Write an equation that describes the Haber process. **1**

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- (c) By analysing the factors affecting ammonia synthesis, explain why the Haber process requires a compromise between these factors. **4**

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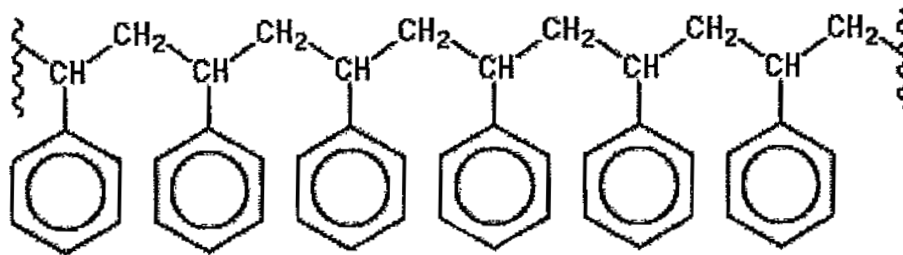
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## Question 27 (4 marks)

Marks

The diagram shows a section of a polymer chain.



- (a) Draw a structure of the monomer from which this polymer is made.

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- (b) State the IUPAC preferred name for this polymer.

1

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- (c) Identify **ONE** common use of this polymer and explain how this use is related to a property of the polymer.

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Page Total

**Question 28** (4 marks)**Marks**

An electrochemical cell is constructed from the following two half cells under standard conditions.

Half cell 1 : a nickel electrode in a solution of 1.0 M nickel nitrate

Half cell 2 : a cadmium electrode in a solution of 1.0 M cadmium nitrate

(a) Draw a labelled diagram of this electrochemical cell. **3**

(b) Given that the standard reduction potential of the cadmium half cell is  $-0.4\text{V}$ , show on your diagram the direction in which electrons will flow in the external circuit of this cell. **1**

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**Question 29** (3 marks)

**Marks**

An electrochemical cell is constructed from the following two half cells under standard conditions.

Half cell 1 : a copper electrode in a solution of 1.0 M copper nitrate

Half cell 2 : a lead electrode in a solution of 1.0 M lead nitrate

- (a) Give the equation for the half reaction that takes place at the anode of this cell. 1

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- (b) Calculate the overall cell voltage of this cell. 2

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## Question 30 (7 marks)

Marks

The following table lists the pH of 0.10 M solutions of four different acids at 25°C.

| Acid | pH  |
|------|-----|
| P    | 1.0 |
| Q    | 3.0 |
| X    | 0.7 |
| Y    | 2.1 |

- (a) Which acid **must** have more than one acidic hydrogen per molecule? Give a reason for your answer.

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- (b) Using the concentration and the pH of acid Y, calculate the percentage ionisation of acid Y in the 0.10 M solution.

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- (c) Samples of the solutions of acids P and Y are diluted by a factor of 10. The resulting **change in pH units** would be:  
(Tick **one** of the following boxes).

1

|                                    |  |
|------------------------------------|--|
| greater for acid P than for acid Y |  |
| greater for acid Y than for acid P |  |
| the same for both acids            |  |

Question 30 continues on page 19

Page Total

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## Question 30 (continued)

Marks

- (d) The dissociation of ethanoic acid in water is exothermic. If a solution of the acid is heated, will the pH of the solution **increase, decrease or remain constant?**

Given an explanation for your answer. Include an equation in your answer

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**Question 31** (7 marks)

**Marks**

- (a) Explain why the presence of ozone in the upper atmosphere is beneficial to life on earth. **1**

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- (b) CFCs can lower the concentration of ozone in the upper atmosphere.

- (i) Which element in the molecules of CFCs causes the destruction of the ozone molecule? **1**

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- (ii) Write an equation which shows how ozone is destroyed by this element. **1**

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- (c) (i) Using electron dot diagrams (Lewis diagrams), draw and label the oxygen molecule and the ozone molecule. Put a circle around the coordinate covalent bond. **2**

- (ii) Referring to their structure and bonding, compare oxygen and ozone over one property. **2**

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### Question 32 (5 marks)

## Marks

“Some bulk water supplies are collected from valleys in wilderness areas, while others are collected from valleys surrounded by mixed farming lands. These different supplies require different levels of clarification and purification before they can be used for human consumption.”

Assess this statement referring to the methods used for treatment of water supplies collected from these different environments.

5

[illegible]

**Question 33** (6 marks)**Marks**

The accuracy of acid-base titrations depends on several factors. These include the primary standard used, how the glassware is prepared and how the equivalence point is determined.

- (a) Explain why sodium hydroxide is not used as a primary standard.

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- (b) During a titration, a conical flask is prepared by rinsing it with distilled water. While this flask is still wet, a clean, dry pipette is used to transfer 20 mL of a standard solution into it. Will the accuracy of the titration be affected? Explain your answer.

**1**

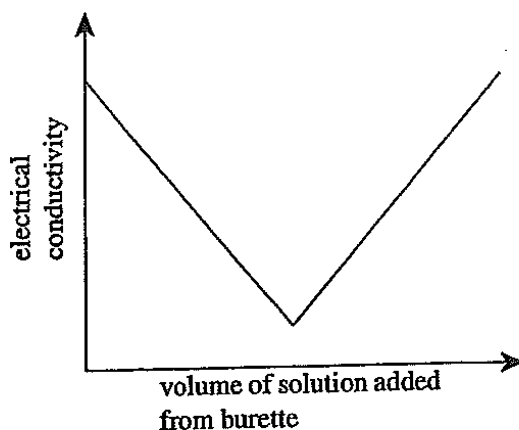
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- (c) Although an indicator can be used to determine the equivalence point of an acid-base titration, an alternative method is to monitor the electrical conductivity of the reaction mixture during the titration. The following graph shows the variation in electrical conductivity during such titration.



Explain why the electrical conductivity:

- (i) starts at a maximum but then decreases to a minimum value.

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Question 33 continues on page 23

Page Total

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**Question 33** (continued)

**Marks**

(ii) does not reach a zero value.

**1**

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(iii) starts to increase again after the minimum value.

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**Question 34** (8 marks)**Marks**

Ethylene is an important resource for chemical industries. It is formed by industrial processes from the components of crude oil (a mixture of alkanes).

- (a) Outline the industrial processes used to convert crude oil into ethylene. Include an equation for the conversion of an alkane into ethylene by one of these industrial processes.

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- (b) Ethanol can be formed from ethylene. Describe how this conversion is achieved and write an equation for the chemical reaction you describe.

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**Question 34 continues on page 25**

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## Question 34 (continued)

Marks

- (c) Ethylene can be distinguished from ethane by a simple laboratory test. Describe how this test is carried out and its results and include balanced equations for chemical reactions which occur in both the dark and in the light.. Name the product formed.

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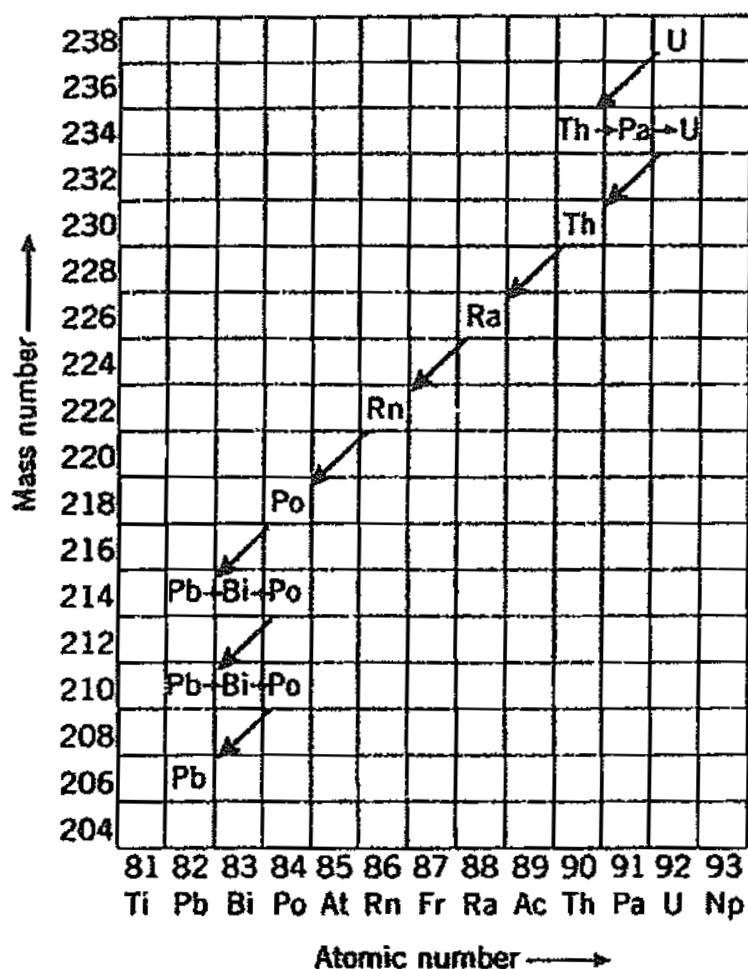
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## Question 35 (2 marks)

Marks

Uranium is a radioactive element that slowly undergoes decay.



Using the flow diagram shown above, trace the sequence of fission products during the decay of uranium for the steps Pa (Protactinium) to Th (Thorium) Th-230.  
(Include the two equations.)

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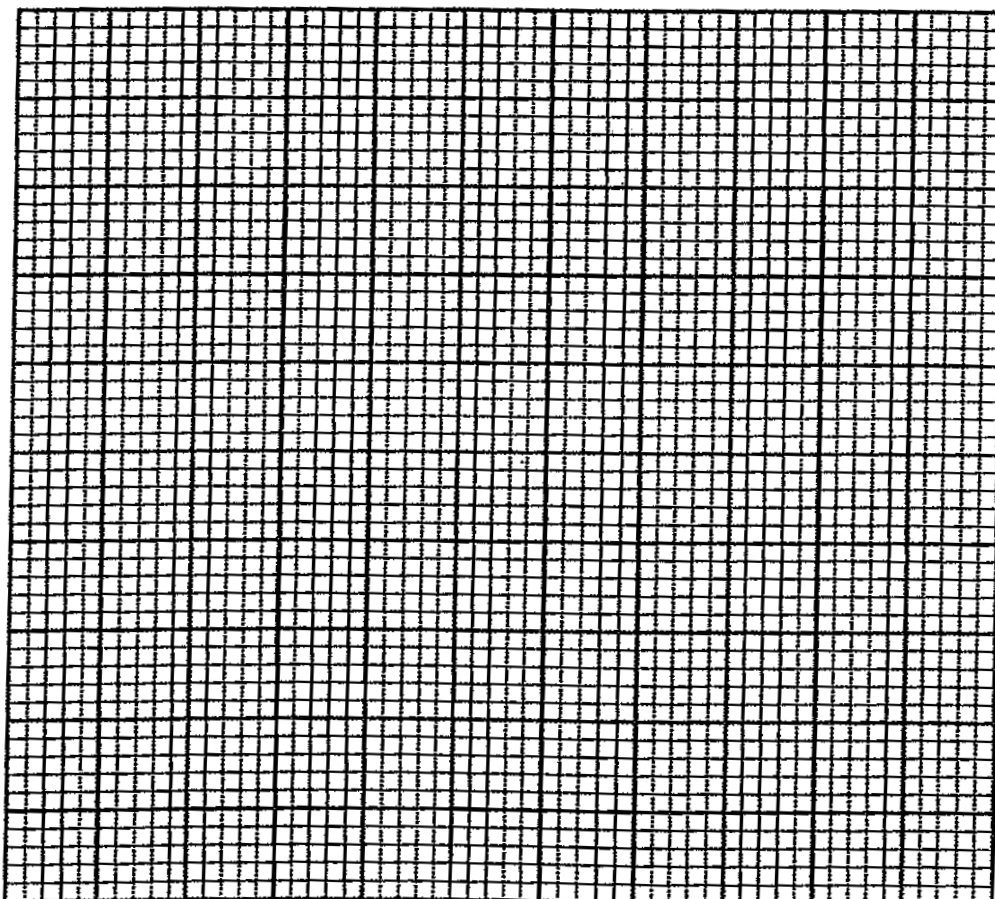
**Question 36** (3 marks)**Marks**

Railways often test the cooling water of their diesel engines for dissolved metals. The presence of particular metal ions in the water gives an indication as to the “health” of the engine. The most commonly used method for this analysis is atomic absorption spectroscopy.

In one particular analysis a chemist was testing for the presence of iron. She used a series of standard solutions to obtain the following results.

| Concentration of Fe (ppm) | Absorbance (%) |
|---------------------------|----------------|
| 0                         | 0.00           |
| 2                         | 0.13           |
| 4                         | 0.28           |
| 6                         | 0.41           |
| 8                         | 0.53           |
| 10                        | 0.67           |

- (a) Draw a calibration graph of these results.

**2**

Question 36 continues on page 28

Page Total



**Question 36** (continued)**Marks**

The chemist then tested the water from three engines.

For each engine she placed 50 mL of sample in a 1000 mL standard flask. The flask was then filled to the mark with distilled, deionised water and shaken to ensure mixing. She then tested each of the diluted samples and obtained the following results.

| Engine Number | Absorbance (%) |
|---------------|----------------|
| X12           | 0.04           |
| X45           | 0.01           |
| X67           | 0.30           |

- (b) Calculate the concentration of iron in the original sample from engine X67.

**1**

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**End of Paper**

Page Total

## DATA SHEET

Avogadro constant,  $N_A$  .....  $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) ..... 22.71 L

at 25°C (298.15 K) ..... 24.79 L

Ionisation constant for water at 25°C (298.15 K),  $K_w$  .....  $1.0 \times 10^{-14}$

Specific heat capacity of water .....  $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

### Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

### Some standard potentials

|                                                                      |                      |                                                  |         |
|----------------------------------------------------------------------|----------------------|--------------------------------------------------|---------|
| $\text{K}^+ + \text{e}^-$                                            | $\rightleftharpoons$ | $\text{K(s)}$                                    | -2.94 V |
| $\text{Ba}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Ba(s)}$                                   | -2.91 V |
| $\text{Ca}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Ca(s)}$                                   | -2.87 V |
| $\text{Na}^+ + \text{e}^-$                                           | $\rightleftharpoons$ | $\text{Na(s)}$                                   | -2.71 V |
| $\text{Mg}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Mg(s)}$                                   | -2.36 V |
| $\text{Al}^{3+} + 3\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Al(s)}$                                   | -1.68 V |
| $\text{Mn}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Mn(s)}$                                   | -1.18 V |
| $\text{H}_2\text{O} + \text{e}^-$                                    | $\rightleftharpoons$ | $\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$  | -0.83 V |
| $\text{Zn}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Zn(s)}$                                   | -0.76 V |
| $\text{Fe}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Fe(s)}$                                   | -0.44 V |
| $\text{Ni}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Ni(s)}$                                   | -0.24 V |
| $\text{Sn}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Sn(s)}$                                   | -0.14 V |
| $\text{Pb}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Pb(s)}$                                   | -0.13 V |
| $\text{H}^+ + \text{e}^-$                                            | $\rightleftharpoons$ | $\frac{1}{2}\text{H}_2(\text{g})$                | 0.00 V  |
| $\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$                       | $\rightleftharpoons$ | $\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$   | 0.16 V  |
| $\text{Cu}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Cu(s)}$                                   | 0.34 V  |
| $\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$ | $\rightleftharpoons$ | $2\text{OH}^-$                                   | 0.40 V  |
| $\text{Cu}^+ + \text{e}^-$                                           | $\rightleftharpoons$ | $\text{Cu(s)}$                                   | 0.52 V  |
| $\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$                       | $\rightleftharpoons$ | $\text{I}^-$                                     | 0.54 V  |
| $\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$                      | $\rightleftharpoons$ | $\text{I}^-$                                     | 0.62 V  |
| $\text{Fe}^{3+} + \text{e}^-$                                        | $\rightleftharpoons$ | $\text{Fe}^{2+}$                                 | 0.77 V  |
| $\text{Ag}^+ + \text{e}^-$                                           | $\rightleftharpoons$ | $\text{Ag(s)}$                                   | 0.80 V  |
| $\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$                      | $\rightleftharpoons$ | $\text{Br}^-$                                    | 1.08 V  |
| $\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$                     | $\rightleftharpoons$ | $\text{Br}^-$                                    | 1.10 V  |
| $\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$        | $\rightleftharpoons$ | $\text{H}_2\text{O}$                             | 1.23 V  |
| $\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$                      | $\rightleftharpoons$ | $\text{Cl}^-$                                    | 1.36 V  |
| $\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$  | $\rightleftharpoons$ | $\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$ | 1.36 V  |
| $\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$                     | $\rightleftharpoons$ | $\text{Cl}^-$                                    | 1.40 V  |
| $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$                         | $\rightleftharpoons$ | $\text{Mn}^{2+} + 4\text{H}_2\text{O}$           | 1.51 V  |
| $\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$                       | $\rightleftharpoons$ | $\text{F}^-$                                     | 2.89 V  |

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

# PERIODIC TABLE OF THE ELEMENTS

|                               | KEY                            |                                | Symbol of element             |                                   | Name of element                    |                                  |                                |                              |
|-------------------------------|--------------------------------|--------------------------------|-------------------------------|-----------------------------------|------------------------------------|----------------------------------|--------------------------------|------------------------------|
|                               | Atomic Number                  | Atomic Weight                  | 79                            | Au                                | 197.0                              | Gold                             |                                |                              |
| 1<br>H<br>1.008<br>Hydrogen   | 4<br>Be<br>9.012<br>Beryllium  | 12<br>Mg<br>24.31<br>Magnesium | 5<br>B<br>10.81<br>Boron      | 6<br>C<br>12.01<br>Carbon         | 7<br>N<br>14.01<br>Nitrogen        | 8<br>O<br>16.00<br>Oxygen        | 9<br>F<br>19.00<br>Fluorine    | 2<br>He<br>4.003<br>Helium   |
| 3<br>Li<br>6.941<br>Lithium   |                                |                                | 13<br>Al<br>26.98<br>Aluminum | 14<br>Si<br>28.09<br>Silicon      | 15<br>P<br>30.97<br>Phosphorus     | 16<br>S<br>32.07<br>Sulfur       | 17<br>Cl<br>35.45<br>Chlorine  | 10<br>Ne<br>20.18<br>Neon    |
| 11<br>Na<br>22.99<br>Sodium   |                                |                                | 31<br>Ga<br>69.72<br>Gallium  | 32<br>Ge<br>72.64<br>Germanium    | 33<br>As<br>74.92<br>Arsenic       | 34<br>Se<br>78.96<br>Selenium    | 35<br>Br<br>79.90<br>Bromine   | 18<br>Ar<br>39.95<br>Argon   |
| 19<br>K<br>39.10<br>Potassium | 20<br>Ca<br>40.08<br>Calcium   |                                | 30<br>Zn<br>65.41<br>Zinc     | 29<br>Cu<br>63.55<br>Copper       | 28<br>Ni<br>58.69<br>Nickel        | 27<br>Co<br>58.93<br>Cobalt      | 26<br>Fe<br>55.85<br>Iron      | 36<br>Kr<br>83.80<br>Krypton |
| 37<br>Rb<br>85.47<br>Rubidium | 38<br>Sr<br>87.62<br>Strontium |                                | 48<br>Cd<br>112.4<br>Cadmium  | 47<br>Ag<br>107.9<br>Silver       | 46<br>Pd<br>106.4<br>Palladium     | 45<br>Rh<br>102.9<br>Rhodium     | 44<br>Ru<br>101.1<br>Ruthenium | 54<br>Xe<br>131.3<br>Xenon   |
| 55<br>Cs<br>132.9<br>Cesium   | 56<br>Ba<br>137.3<br>Barium    |                                | 80<br>Hg<br>200.6<br>Mercury  | 79<br>Au<br>197.0<br>Gold         | 78<br>Pt<br>195.1<br>Platinum      | 77<br>Ir<br>192.2<br>Iridium     | 76<br>Os<br>190.2<br>Osmium    | 86<br>Rn<br>[222.0]<br>Radon |
| 87<br>Fr<br>[223]<br>Francium | 88<br>Ra<br>[226]<br>Radium    |                                |                               | 111<br>Rg<br>[272]<br>Roentgenium | 110<br>Ds<br>[271]<br>Darmstadtium | 109<br>Mt<br>[268]<br>Meitnerium | 108<br>Hs<br>[277]<br>Hassium  |                              |
|                               |                                |                                |                               |                                   |                                    |                                  |                                |                              |

## Lanthanoids

|                                |                             |                                   |                                |                                 |                               |                               |                                 |                              |                                 |                              |                             |                              |                                |                               |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|---------------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|
| 57<br>La<br>138.9<br>Lanthanum | 58<br>Ce<br>140.1<br>Cerium | 59<br>Pr<br>140.9<br>Praseodymium | 60<br>Nd<br>144.2<br>Neodymium | 61<br>Pm<br>[145]<br>Promethium | 62<br>Sm<br>150.4<br>Samarium | 63<br>Eu<br>152.0<br>Europium | 64<br>Gd<br>157.3<br>Gadolinium | 65<br>Tb<br>158.9<br>Terbium | 66<br>Dy<br>162.5<br>Dysprosium | 67<br>Ho<br>164.9<br>Holmium | 68<br>Er<br>167.3<br>Erbium | 69<br>Tm<br>168.9<br>Thulium | 70<br>Yb<br>173.0<br>Ytterbium | 71<br>Lu<br>175.0<br>Lutetium |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|---------------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|

## Actinoids

|                               |                              |                                   |                             |                                |                                |                                |                             |                                |                                  |                                  |                               |                                   |                                |                                  |
|-------------------------------|------------------------------|-----------------------------------|-----------------------------|--------------------------------|--------------------------------|--------------------------------|-----------------------------|--------------------------------|----------------------------------|----------------------------------|-------------------------------|-----------------------------------|--------------------------------|----------------------------------|
| 89<br>Ac<br>[227]<br>Actinium | 90<br>Th<br>232.0<br>Thorium | 91<br>Pa<br>231.0<br>Protactinium | 92<br>U<br>238.0<br>Uranium | 93<br>Np<br>[237]<br>Neptunium | 94<br>Pu<br>[244]<br>Plutonium | 95<br>Am<br>[243]<br>Americium | 96<br>Cm<br>[247]<br>Curium | 97<br>Bk<br>[247]<br>Berkelium | 98<br>Cf<br>[251]<br>Californium | 99<br>Es<br>[252]<br>Einsteinium | 100<br>Fm<br>[257]<br>Fermium | 101<br>Md<br>[258]<br>Mendelevium | 102<br>No<br>[259]<br>Nobelium | 103<br>Lr<br>[262]<br>Lawrencium |
|-------------------------------|------------------------------|-----------------------------------|-----------------------------|--------------------------------|--------------------------------|--------------------------------|-----------------------------|--------------------------------|----------------------------------|----------------------------------|-------------------------------|-----------------------------------|--------------------------------|----------------------------------|

For elements that have no stable or long-lived nuclides, the mass number of the nuclide with the longest confirmed half-life is listed between square brackets.  
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (October 2005 version) is the principal source of data. Some data may have been modified.



FINAL PRINTED VERSION

23 / 7 / 10

Rob Paynter

Student Number

RJP  
SUGGESTED ANSWERS

Barker College

2010

TRIAL  
HIGHER SCHOOL  
CERTIFICATE

## Chemistry

Staff Involved:

AM WEDNESDAY 4<sup>TH</sup> AUGUST

- RJP
- RZS \*
- KHW
- NJD

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Section I

Multiple Choice

Choose the best response and fill in the response oval completely

SECTION I  
ANSWER SHEET

|     |                                  |                                  |                                  |                                  |
|-----|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| 1.  | <input type="radio"/> A          | <input type="radio"/> B          | <input type="radio"/> C          | <input checked="" type="radio"/> |
| 2.  | <input type="radio"/> A          | <input checked="" type="radio"/> | <input type="radio"/> C          | <input type="radio"/> D          |
| 3.  | <input type="radio"/> A          | <input checked="" type="radio"/> | <input type="radio"/> C          | <input type="radio"/> D          |
| 4.  | <input checked="" type="radio"/> | <input type="radio"/> B          | <input type="radio"/> C          | <input type="radio"/> D          |
| 5.  | <input type="radio"/> A          | <input type="radio"/> B          | <input type="radio"/> C          | <input checked="" type="radio"/> |
| 6.  | <input type="radio"/> A          | <input type="radio"/> B          | <input checked="" type="radio"/> | <input type="radio"/> D          |
| 7.  | <input type="radio"/> A          | <input type="radio"/> B          | <input type="radio"/> C          | <input checked="" type="radio"/> |
| 8.  | <input type="radio"/> A          | <input type="radio"/> B          | <input checked="" type="radio"/> | <input type="radio"/> D          |
| 9.  | <input type="radio"/> A          | <input checked="" type="radio"/> | <input type="radio"/> C          | <input type="radio"/> D          |
| 10. | <input type="radio"/> A          | <input checked="" type="radio"/> | <input type="radio"/> C          | <input type="radio"/> D          |
| 11. | <input type="radio"/> A          | <input type="radio"/> B          | <input checked="" type="radio"/> | <input type="radio"/> D          |
| 12. | <input type="radio"/> A          | <input type="radio"/> B          | <input checked="" type="radio"/> | <input type="radio"/> D          |
| 13. | <input type="radio"/> A          | <input checked="" type="radio"/> | <input type="radio"/> C          | <input type="radio"/> D          |
| 14. | <input checked="" type="radio"/> | <input type="radio"/> B          | <input type="radio"/> C          | <input type="radio"/> D          |
| 15. | <input type="radio"/> A          | <input checked="" type="radio"/> | <input type="radio"/> C          | <input type="radio"/> D          |
| 16. | <input type="radio"/> A          | <input type="radio"/> B          | <input checked="" type="radio"/> | <input type="radio"/> D          |
| 17. | <input type="radio"/> A          | <input checked="" type="radio"/> | <input type="radio"/> C          | <input type="radio"/> D          |
| 18. | <input checked="" type="radio"/> | <input type="radio"/> B          | <input type="radio"/> C          | <input type="radio"/> D          |
| 19. | <input type="radio"/> A          | <input type="radio"/> B          | <input type="radio"/> C          | <input checked="" type="radio"/> |
| 20. | <input type="radio"/> A          | <input type="radio"/> B          | <input type="radio"/> C          | <input checked="" type="radio"/> |



# Barker College

## 2010 TRIAL HIGHER SCHOOL CERTIFICATE

# Chemistry

Staff Involved:

AM WEDNESDAY 4<sup>TH</sup> AUGUST

- RJP
- RZS \*
- KHW
- NJD

94 copies

### General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- Draw diagrams using pencil
- A Data Sheet and Periodic Table are provided at the back of this paper
- Write your Barker Student Number at the top of the Section I Answer Sheet, at the top of this page and at the top of ALL answer pages in Section II
- Show ALL working for calculations

Total marks – 100

### Section I

Pages 2 - 7

20 marks

- Multiple Choice
- Indicate all answers on the Answer Sheet provided
- Allow about 30 minutes for this section.

### Section II

Pages 8 - 28

80 marks

- Short Response
- Indicate all answers in the spaces provided on the paper
- Allow about 150 minutes for this section

**Section I**  
**20 marks**

**Attempt Questions 1-20**

**Allow about 30 minutes for this part**

Use the multiple-choice answer sheet provided for Questions 1-20.

---

1. Which of the following is classified as a biopolymer?
  - (A) PVC
  - (B) Polystyrene
  - (C) Glucose
  - (D) Cellulose
  
2. Which of the following species has the greatest ability to gain electrons from a reducing agent in solution?
  - (A) Zinc ions
  - (B) Silver ions
  - (C) Zinc
  - (D) Silver
  
3. Which of the following elements is classified as transuranic?
  - (A) Cerium
  - (B) Curium
  - (C) Caesium
  - (D) Cadmium
  
4. When glucose is fermented to form ethanol and carbon dioxide, the carbon dioxide produced can be bubbled into limewater,  $\text{Ca}(\text{OH})_2$  to form a white precipitate of calcium carbonate. What is the maximum mass of calcium carbonate which could be formed from 90 g of glucose?
  - (A) 100 g
  - (B) 50 g
  - (C) 200 g
  - (D) 90 g

Section I continued

5. Some salts, when dissolved in water, produce acidic or alkaline solutions, while others produce neutral solutions. Which is the CORRECT statement?
- (A) When sodium chloride is dissolved in water, the pH increases slightly.
  - (B) When potassium ethanoate is dissolved in water, the concentration of hydroxide ions decreases.
  - (C) When sodium carbonate is dissolved in water, the pH decreases.
  - (D) When ammonium nitrate is dissolved in water, the concentration of hydronium ions increases.
6. An indication that an oxide is classified as basic is when it:
- (A) reacts with water to form a base or with a salt to form an acid.
  - (B) reacts with water to form an acid or with a base to form a salt.
  - (C) reacts with an acid to form a salt or with water to form hydroxide ions.
  - (D) turns litmus paper red.
7. Which of the following mixtures would form a buffer solution?
- (A) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOOH}$  and 1000 mL of  $0.1 \text{ mol L}^{-1} \text{CH}_3\text{COO}^- \text{Na}^+$
  - (B) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HNO}_3$  and 25 mL of  $1 \text{ mol L}^{-1} \text{NO}_3^-$
  - (C) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOOH}$  and 100 mL of  $0.1 \text{ mol L}^{-1} \text{CH}_3\text{COOH}$
  - (D) 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOOH}$  and 100 mL of  $0.1 \text{ mol L}^{-1} \text{HCOO}^- \text{Na}^+$
8. The chemical added to bulk water supplies to make the solution alkaline is:
- (A) chlorine
  - (B) sodium fluoride
  - (C) lime
  - (D) iron (III) chloride

**Section I continued**

9. Some measurements were made on various water samples and the results tabulated.

| Sample                   | A   | B   | C   | D   |
|--------------------------|-----|-----|-----|-----|
| DO (ppm)                 | 2.2 | 7.0 | 8.7 | 5.6 |
| TDS (ppm)                | 550 | 600 | 45  | 200 |
| Turbidity (NTU)          | 100 | 250 | 5   | 10  |
| pH                       | 7.5 | 3.5 | 6.5 | 7.5 |
| Coliforms (CFU / 100 mL) | 200 | 0   | 0   | 0   |

The sample most likely to have been drawn from a water supply polluted by run-off from a mining site is:

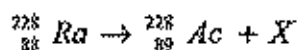
- (A) A  
(B) B  
(C) C  
(D) D
10. Which of the following is the conjugate acid of  $H_2PO_4^-$ ?
- (A)  $H_2PO_4$   
(B)  $H_3PO_4$   
(C)  $HPO_4^{2-}$   
(D)  $PO_4^{3-}$
11. Consider the four chloroalkanes
- 2, 4-dichloropentane
  - 2, 4-dichlorohexane
  - 2, 3-dichlorohexane
  - 2, 2, 3, 4-tetrachloropentane

Which of the four chloroalkanes are isomers?

- (A) 1 and 2  
(B) 1 and 4  
(C) 2 and 3  
(D) 3 and 4

**Section I continued**

12. Atomic absorption spectroscopy (AAS) would be a suitable method for determining the concentration of:
- (A) Phosphorus in detergents
  - (B) Ethanol in wine
  - (C) Mercury in fish
  - (D) Nitrogen in fertilisers
13. 0.1 mol/L citric acid is neutralised by a solution of 0.1 mol/L sodium hydroxide. Choose the best indicator for this titration.
- (A) Methyl orange
  - (B) Phenolphthalein
  - (C) Bromothymol blue
  - (D) Universal
14. Butanoic acid and ethyl butanoate have the same molecular formula,  $C_6H_{12}O_2$ . Which of the following is the best chemical test to distinguish between these two compounds?
- (A) Dilute sodium carbonate
  - (B) Bromine water
  - (C) Solubility in water
  - (D) Smell
15. The following equation represents the nuclear decay of radium - 88.



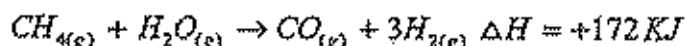
Identify the species marked X.

- (A) An alpha particle
- (B) A beta particle
- (C) A neutron
- (D) A proton

**Section I continued**

16. The quantity of flocculent added to a water supply was less than the amount required to effectively treat that supply. What would be the effect on the water quality?
- (A) A high value of TDS (total dissolved solids)
  - (B) A pH between 5 and 6
  - (C) Turbid water
  - (D) A high bacterial count

17. One possible source of hydrogen for use of a fuel is through the "reforming" of natural gas. The key reaction is :



A catalyst is used for this reaction.

Which of the following is the most suitable set of conditions for this reaction?

- (A) Low pressure and low temperature
  - (B) Low pressure and high temperature
  - (C) High pressure and high temperature
  - (D) High pressure and low temperature
18. In an aqueous solution of the weak acid nitrous acid,  $HNO_2$ , which of the following species is present? IN THE HIGHEST CONCENTRATION.
- (A)  $HNO_2$
  - (B)  $H_3O^+$
  - (C)  $OH^-$
  - (D)  $NO_2^-$
19. Which of the following reactions will occur as written?
- (A)  $Cu_{(s)} + Zn(NO_3)_{2(aq)} \rightarrow Cu(NO_3)_{2(aq)} + Zn_{(s)}$
  - (B)  $Cu_{(s)} + Sn(NO_3)_{2(aq)} \rightarrow Cu(NO_3)_{2(aq)} + Sn_{(s)}$
  - (C)  $2Ag_{(s)} + Cu(NO_3)_{2(aq)} \rightarrow 2AgNO_{3(aq)} + Cu_{(s)}$
  - (D)  $Sn_{(s)} + Cu(NO_3)_{2(aq)} \rightarrow Sn(NO_3)_{2(aq)} + Cu_{(s)}$

**Section I continued**

20. Why is the maximum yield of ethanol produced by the yeast - catalysed fermentation of 1 litre of glucose solution approximately 150ml?
- (A) The reaction is limited by the amount of water.
  - (B) There is too much heat generated if more ethanol is produced.
  - (C) Too much oxygen is generated if more ethanol is produced.
  - (D) The yeast can only tolerate an environment with a maximum 15% alcohol.

**End of Section I**

Section II

Total marks 80

Attempt Question 21 - 36

Allow about 150 minutes for this section.

P 8-12 RJP 20  
P 13-17 KHW 22  
P 18-21 HJD 19  
P 22-28 RZS 19.

Student Number: .....

Question 21 (3 marks)

Marks

A student prepared a sample of an indicator, using red cabbage. She tested the indicator with solutions of known pH and tabulated her observations.

| Colour | Red |   | violet |   | purple |   | blue |   | green |    | yellow |    |    |    |
|--------|-----|---|--------|---|--------|---|------|---|-------|----|--------|----|----|----|
| pH     | 1   | 2 | 3      | 4 | 5      | 6 | 7    | 8 | 9     | 10 | 11     | 12 | 13 | 14 |

- (a) State the colour of the indicator in a  $0.0005 \text{ mol L}^{-1}$  solution of sulfuric acid. Show your working.

2

$$[\text{H}_2\text{SO}_4] = 0.0005 \text{ mol. L}^{-1} \therefore [\text{H}^+] = 0.001 \text{ mol. L}^{-1}$$

$$\text{pH} = -\log [\text{H}^+]$$

$$\text{pH} = -\log 1 \times 10^{-3}$$

$$\text{pH} = 3 \therefore \text{COLOUR IS VIOLET}$$

- (b) 1.00 mL of the  $0.0005 \text{ mol L}^{-1}$  solution of sulfuric acid was diluted to 1 L, using deionised water. What could be the colour of the diluted solution after addition of a few drops of red cabbage indicator? Show working.

1

$$C_1 V_1 = C_2 V_2$$

$$0.0005 \times 1 = C_2 \times 1000$$

$$C_2 = 5 \times 10^{-7} \text{ mol. L}^{-1} \text{ for } \text{H}_2\text{SO}_4$$

$$\therefore [\text{H}_3\text{O}^+] = 1 \times 10^{-6} \text{ M}$$

$$\text{pH} = 6$$

COLOUR IS PURPLE.

Page Total

## Question 22 (5 marks)

Marks

Cobalt-60 (Co - 60) is a gamma emitter. It is produced by the neutron bombardment of Co-59.

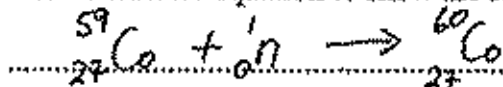
- (a) Name an instrument that can be used to detect gamma radiation.

1

GEIGER COUNTER, SCINTILLATION COUNTER, CLOUD CHAMBER.

- (b) Write a nuclear equation to show the formation of Co-60 from Co - 59.

1



- (c) Evaluate the medical use of Co-60 or another identified medical radioisotope you have studied.

3

Co-60 emits GAMMA RAY which carry a suitable amount of energy to DESTROY CANCER CELLS

Co-60 has a suitable HALF-LIFE (4 to 6 YEARS) so radiation source has a reasonable lifetime in the equipment.

Despite the possibility of <sup>some</sup> good cells being destroyed during treatment Co-60 remains a valuable medical radioisotope for treating cancer.

STERILISE MEDICAL EQUIPMENT

TECHNETIUM 99m - NON-INVASIVE and HIGHLY SENSITIVE means of DIAGNOSIS of conditions such as OSTEOMYELITIS. HALF-LIFE OF 6 HOURS and is a GAMMA EMITTER. Short half-life means the EXPOSURE of patient and doctor to radiation is MINIMISED.

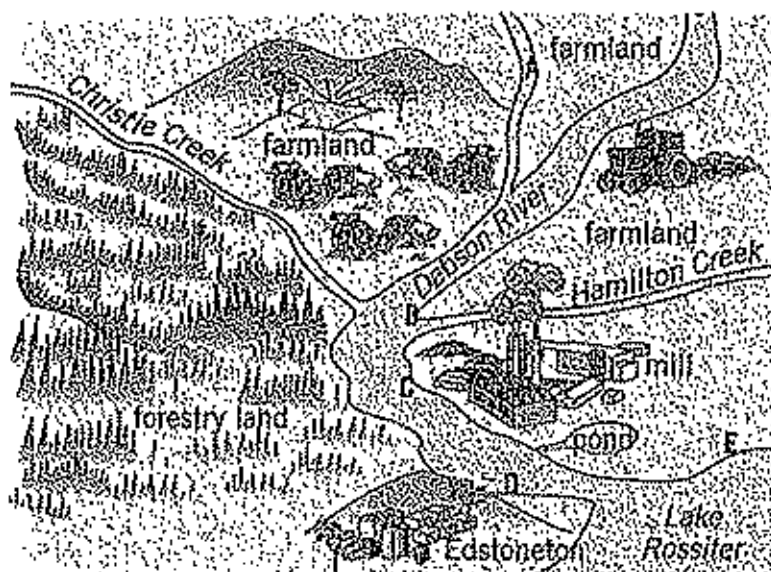
Despite these intrinsic risks of production in a nuclear reactor, TRANSPORTATION and STORAGE, the benefits of NON-INVASIVE and ACCURATE DIAGNOSIS OUTWEIGH the possible risks when appropriate SAFETY PRECAUTIONS are taken.

Page Total

## Question 23 (7 marks)

Marks

The map below shows the location of several sites which affect the water quality of the town, Edstoneton. Various water testing stations are labelled A - E.



- (i) The water temperature at testing station D is quite high. Account for this high temperature.

1

Mill could be using water to cool plant and return warmed water to the river. WARM WATER FROM TOWN (1/2)  
HIGH TURBIDITY leads to ↑ T (1/2)

- (ii) The water in the Debson River experiences algal blooms especially after heavy rain. Name the possible nutrient responsible for the algal bloom and state its likely source. 2

NITRATE OR PHOSPHATE from farmland.

PHOSPHORUS (1/2)

- (iii) What effect could increased water temperature have on the level of dissolved oxygen in the river? Explain your answer.

2

Level of DISSOLVED OXYGEN would DECREASE ✓  
The SOLUBILITY OF GASES DECREASES with INCREASING TEMPERATURE.

1/2 if increased BACTERIA etc

Question 23 continues on page 11

Page Total

|  |
|--|
|  |
|--|

## Question 23 (continued)

Marks

- (iv) Recently some of the forestry land along Christie Creek has been cleared for urban development. This has affected the water quality at Testing Station B particularly after heavy rain. What would an observer notice about a water sample at Testing Station B?

1

HIGH TURBIDITY.

- (v) Describe ONE water treatment method that could be used to sterilise the water at site A so that it would be suitable for drinking.

1

GASEOUS CHLORINE could be dissolved in it. This destroys bacteria and some viruses.

MEMBRANE FILTERS

OZONE.

DISTILLATION.

HIGH UV

BOILING

$\frac{1}{2}$  if just listed  
all methods.

$\frac{1}{2}$  if just said CHLORINATION  
but did not DESCRIBE.

Page Total

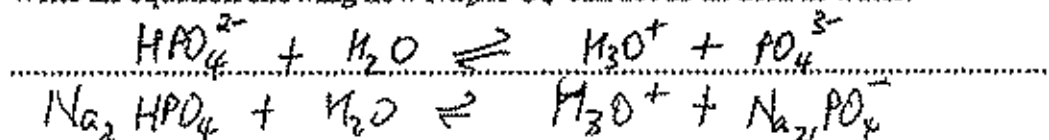
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## Question 24 (5 marks)

Sodium hydrogen phosphate ( $\text{Na}_2\text{HPO}_4$ ) forms an amphiprotic species in water that is involved in the buffering of living cells.

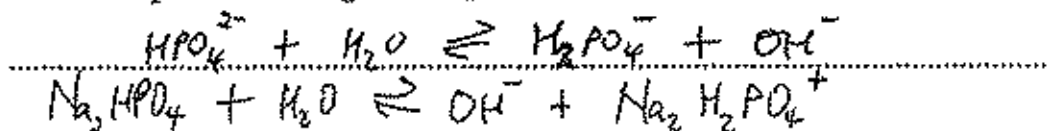
- (a) Write an equation showing how  $\text{Na}_2\text{HPO}_4$  can act as an acid in water.

1



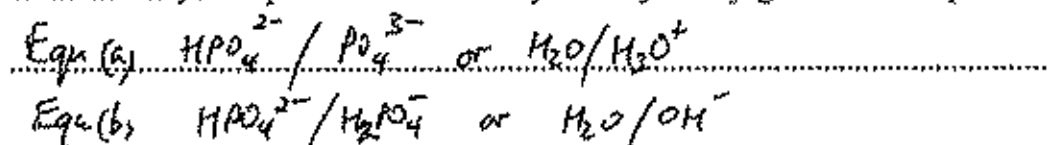
- (b) Write an equation showing how  $\text{Na}_2\text{HPO}_4$  can act as a base in water.

1



- (c) From one of your equations above clearly identify a conjugate acid-base pair.

1



- (d) The behaviour of the above amphiprotic species ( $\text{HPO}_4^{2-}$ ) fits the definition of an acid / base theory you have studied. Identify the theory and explain how it fits that definition.

2

LOWRY-BRONSTED

In equation (a)  $\text{HPO}_4^{2-}$  is acting as a PROTON DONOR. In equation (b)  $\text{HPO}_4^{2-}$  is acting as a PROTON ACCEPTOR.

## Question 25 (4 marks)

In class you analysed, by means of titration, a sample of fertiliser to determine the percentage of sulfate it contained. Below is a typical table of results.

| Titre | BaCl <sub>2</sub><br>Volume (mL) |
|-------|----------------------------------|
| 1     | 29.50                            |
| 2     | 29.45                            |
| 3     | 29.42                            |

9.32 g of the original fertiliser were dissolved in 250 mL of deionised water. 10 mL of this solution were pipetted into a conical flask. The burette contained 0.051 mol L<sup>-1</sup> barium chloride.

Calculate the percentage of sulfate in the fertiliser. Show all your working as a means of explaining the chemistry involved.

4

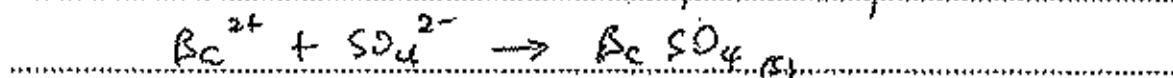
$$\text{Average titre} = \frac{29.50 + 29.45 + 29.42}{3} = 29.46 \text{ mL}$$

$$\text{moles of BaCl}_2 \quad n = cV$$

$$n = (0.051) \times (29.46 \times 10^{-3})$$

$$n = 1.50246 \times 10^{-3} \text{ mol}$$

$$\therefore \text{same number of moles of Ba}^{2+}$$



$$\text{moles of SO}_4^{2-} \text{ is also } 1.50246 \times 10^{-3} \text{ mol in 10 mL}$$

$$\text{moles of SO}_4^{2-} \text{ in 250 mL} = (1.50246 \times 10^{-3}) \times (25)$$

$$= 0.03756 \text{ mol}$$

$$\text{mass of SO}_4^{2-} \text{ in 250 mL} = (0.03756) \times (96.07)$$

$$= 3.609 \text{ g}$$

$$\% \text{ SO}_4 = \frac{3.609}{9.32} \times \frac{100}{1}$$

$$= 38.72 \%$$

## Question 26 (7 marks)

Marks

The Haber process was developed in 1908 as a method of synthesising ammonia and was converted into a successful industrial process during the following decade.

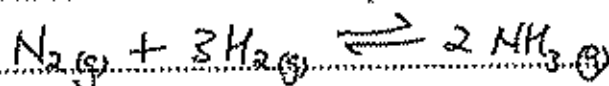
- (a) Evaluate the significance of the Haber process at that time in world history.

2

World War I was occurring at that time. The British had a NAVAL BLOCKADE of German supply ships bringing <sup>SALT PETER</sup> nitrate from CHILE. The HABER process enabled Germany to have a supply of  $\text{NH}_3$ , despite the blockade. The  $\text{NH}_3$  could then be used to make EXPLOSIVES and FERTILISERS. The HABER PROCESS was highly significant as it PROLONGED the war effort by GERMANY.

- (b) Write an equation that describes the Haber process.

1



- (c) By analysing the factors affecting ammonia synthesis, explain why the Haber process requires a compromise between these factors.

4

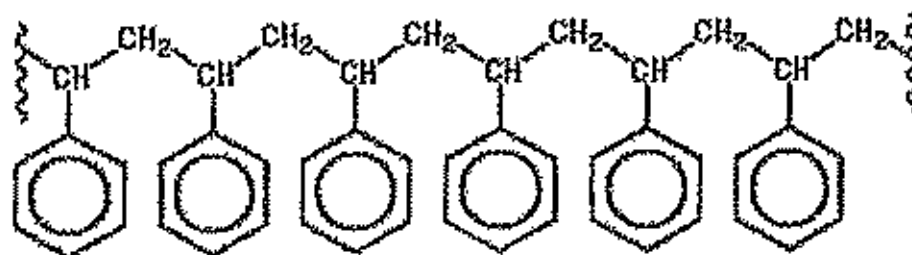
PRESSURE: According to Le Chatelier's Principle the higher the pressure the more  $\text{NH}_3$  produced. However high pressures require VERY EXPENSIVE equipment and there are SAFETY ISSUES. A compromise pressure of 250 ATM is used.

TEMPERATURE: The HABER PROCESS is EXOTHERMIC releasing 92 kJ mol<sup>-1</sup> of  $\text{H}_2$  reacted. According to Le Chatelier's Principle if we LOWER THE TEMPERATURE at equilibrium then the reaction which releases energy will be favoured and this is the forward reaction producing  $\text{NH}_3$ . However at low temperatures REACTION RATE is too slow so a compromise temperature of about 400°C is used.

## Question 27 (4 marks)

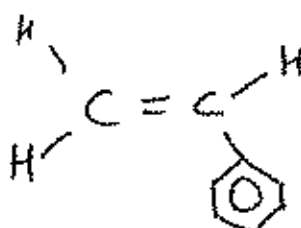
Marks

The diagram shows a section of a polymer chain.



- (a) Draw a structure of the monomer from which this polymer is made.

1



- (b) State the IUPAC preferred name for this polymer.

1

POLYSTYRENE

- (c) Identify ONE common use of this polymer and explain how this use is related to a property of the polymer.

2

TOOL HANDLES:

CD CASES

CAR BATTERY CASES

Polystyrene is HARD, STRONG PLASTIC

Page Total

## Question 28 (4 marks)

Marks

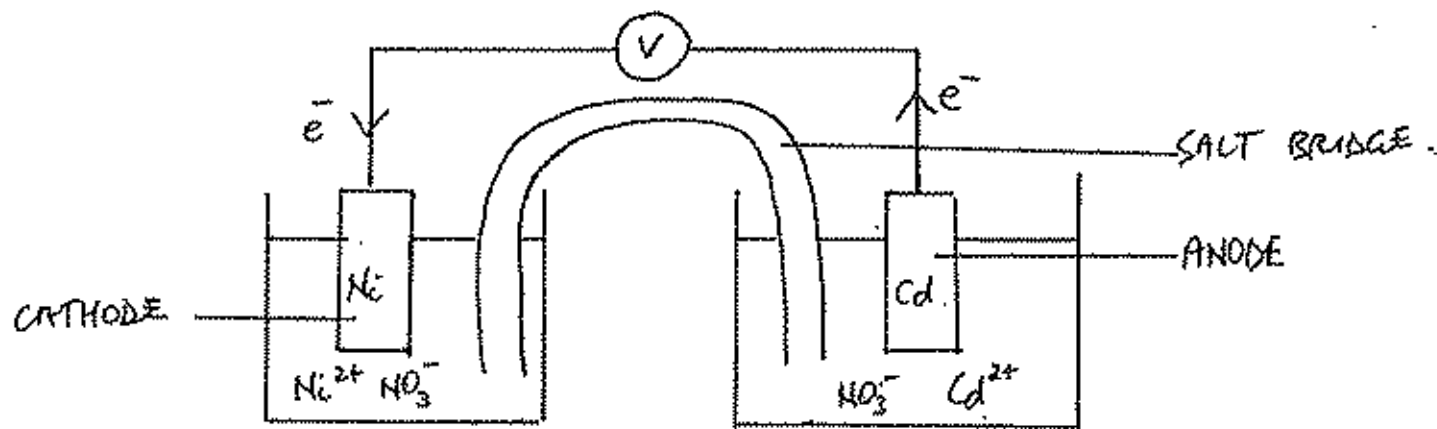
An electrochemical cell is constructed from the following two half cells under standard conditions.

Half cell 1 : a nickel electrode in a solution of 1.0 M nickel nitrate

Half cell 2 : a cadmium electrode in a solution of 1.0 M cadmium nitrate

- (a) Draw a labelled diagram of this electrochemical cell.

3



- (b) Given that the standard reduction potential of the cadmium half cell is  $-0.4\text{V}$ , show on your diagram the direction in which electrons will flow in the external circuit of this cell.

1

## Question 29 (3 marks)

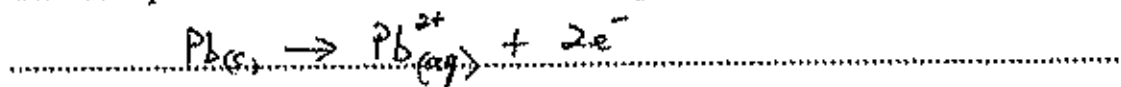
Marks

An electrochemical cell is constructed from the following two half cells under standard conditions.

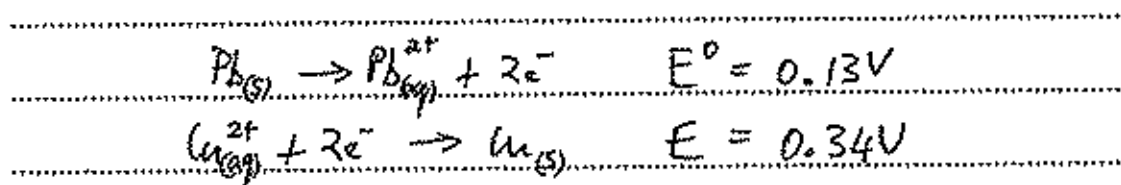
Half cell 1 : a copper electrode in a solution of 1.0 M copper nitrate

Half cell 2 : a lead electrode in a solution of 1.0 M lead nitrate

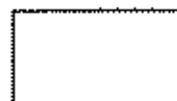
- (a) Give the equation for the half reaction that takes place at the anode of this cell. 1



- (b) Calculate the overall cell voltage of this cell. 2



$$\begin{aligned} \text{EMF} &= E_{\text{ox}} + E_{\text{red}} \\ \text{EMF} &= 0.13\text{V} + 0.34\text{V} \\ \text{EMF} &= 0.47\text{V} \end{aligned}$$



## Question 30 (7 marks)

Marks

The following table lists the pH of 0.10 M solutions of four different acids at 25°C.

| Acid | pH  |
|------|-----|
| P    | 1.0 |
| Q    | 3.0 |
| X    | 0.7 |
| Y    | 2.1 |

- (a) Which acid must have more than one acidic hydrogen per molecule? Give a reason for your answer.

2

Acid X. If acid X was monoprotic it would have a pH of 1. Since the pH is less than 1.0, the  $[H^+]$  must be greater than 0.1 M and the acid must be diprotic at least.

- (b) Using the concentration and the pH of acid Y, calculate the percentage ionisation of acid Y in the 0.10 M solution.

1

$pH = 2.1$ ,  $[H^+] = 10^{-2.1}$ ,  $[H^+] = 7.94 \times 10^{-3}$   
 Fully ionised  $[H^+] = 0.1$   $\therefore \% \text{ ionisation} = \frac{7.94 \times 10^{-3}}{0.1} \times \frac{100}{1}$   
 $= 7.94\%$

- (c) Samples of the solutions of acids P and Y are diluted by a factor of 10.

The resulting change in pH units would be:

1

(Tick one of the following boxes).

|                                    |   |
|------------------------------------|---|
| greater for acid P than for acid Y |   |
| greater for acid Y than for acid P |   |
| the same for both acids            | ✓ |

Question 30 continues on page 19

Page Total

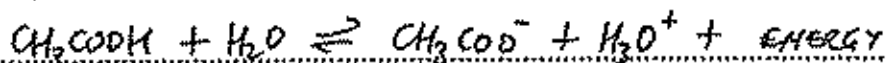
## Question 30 (continued)

Marks

- (d) The dissociation of ethanoic acid in water is exothermic. If a solution of the acid is heated, will the pH of the solution increase, decrease or remain constant?

Give an explanation for your answer. Include an equation in your answer

3



According to Le Chatelier heating the system will favour the REVERSE ENDOTHERMIC reaction as it ABSORBS ENERGY.

The  $[\text{H}_3\text{O}^+]$  would DECREASE  $\therefore$  the pH INCREASES.

## Question 31 (7 marks)

Marks

- (a) Explain why the presence of ozone in the upper atmosphere is beneficial to life on earth. 1

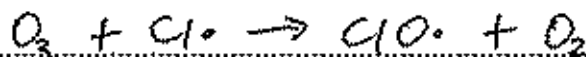
OZONE ABSORBS HARMFUL U-V RADIATION.This radiation if not absorbed would cause increased incidence of SKIN CANCER, EYE CATARACTS

- (b) CFCs can lower the concentration of ozone in the upper atmosphere.

- (i) Which element in the molecules of CFCs causes the destruction of the ozone molecule? 1

CHLORINE

- (ii) Write an equation which shows how ozone is destroyed by this element. 1

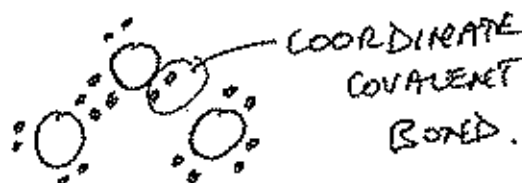


- (c) (i) Using electron dot diagrams (Lewis diagrams), draw and label the oxygen molecule and the ozone molecule. Put a circle around the coordinate covalent bond. 2

OXYGEN



OZONE



- (ii) Referring to their structure and bonding, compare oxygen and ozone over one property. 2

Melting or boiling point.

O<sub>2</sub> is NON-POLAR i.e. ONLY DISPERSION FORCES between molecules. O<sub>3</sub> is POLAR i.e. has DIPOLE/DIPOLE FORCES as well as DISPERSION FORCES hence O<sub>3</sub> has HIGHER m.p. and b.p. compared to O<sub>2</sub>



## Question 32 (5 marks)

Marks

"Some bulk water supplies are collected from valleys in wilderness areas, while others are collected from valleys surrounded by mixed farming lands. These different supplies require different levels of clarification and purification before they can be used for human consumption."

Assess this statement referring to the methods used for treatment of water supplies collected from these different environments.

5

Clarification involves removing any turbidity or colour so that the water is sparklingly clear and colourless. Water that is collected in wilderness areas needs far less clarification than that collected from valleys surrounded by farming lands. Farming land water will need lime added to raise the pH then  $Fe^{3+}$  ions are added to form a precipitate and enable COAGULATION to occur. This takes most of the SUSPENDED matter out of the water which is then FILTERED through sand.

Water from both sources will need to be sanitised. This can be done by bubbling chlorine through the water to kill bacteria, or membrane filters could be used, but they are very expensive.

The statement is accurate as far more clarifying is needed in the case of the farmland water.



## Question 33 (6 marks)

Marks

The accuracy of acid-base titrations depends on several factors. These include the primary standard used, how the glassware is prepared and how the equivalence point is determined.

- (a) Explain why sodium hydroxide is not used as a primary standard.

1

Its concentration does not remain constant as NaOH absorbs water from the atmosphere

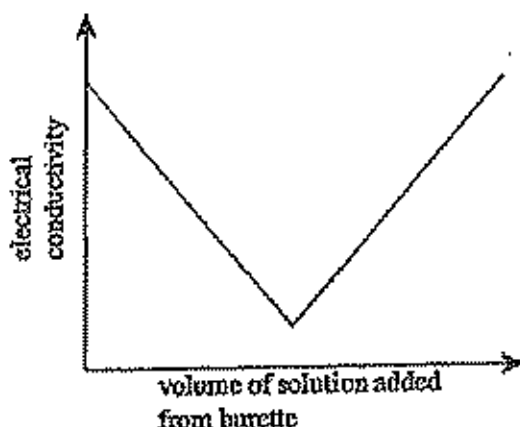
- (b) During a titration, a conical flask is prepared by rinsing it with distilled water. While this flask is still wet, a clean, dry pipette is used to transfer 20 mL of a standard solution into it. Will the accuracy of the titration be affected? Explain your answer.

1

NO:

The NUMBER OF MOLES of standard solution has not been changed by having some water in the conical flask.

- (c) Although an indicator can be used to determine the equivalence point of an acid-base titration, an alternative method is to monitor the electrical conductivity of the reaction mixture during the titration. The following graph shows the variation in electrical conductivity during such titration.



Explain why the electrical conductivity:

- (i) starts at a maximum but then decreases to a minimum value.

2

Initially there is a maximum number of IONS, high conductivity. As salt and water form IONS are REMOVED SO CONDUCTIVITY DECREASES.

Question 33 continues on page 23

Page Total

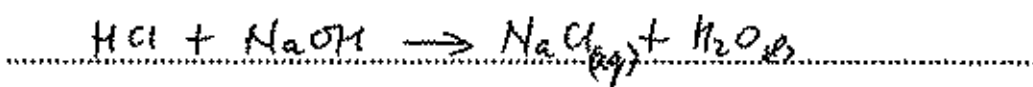
## Question 33 (continued)

Marks

- (ii) does not reach a zero value.

1

THERE ARE STILL IONS IN SOLUTION (DISSOLVED NaCl).



- (iii) starts to increase again after the minimum value.

1

The extra acid (OR BASE) that is added will provide more IONS so conductivity will INCREASE.



## Question 34 (8 marks)

Marks

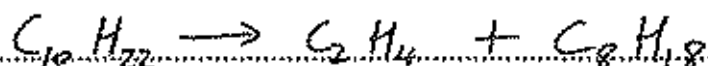
Ethylene is an important resource for chemical industries. It is formed by industrial processes from the components of crude oil (a mixture of alkanes).

- (a) Outline the industrial processes used to convert crude oil into ethylene. Include an equation for the conversion of an alkane into ethylene by one of these industrial processes.

3

FRACTIONAL DISTILLATION: Crude oil is heated in a fractionating column and various hydrocarbons CONDENSE AT SPECIFIC TEMPERATURES and can be collected.

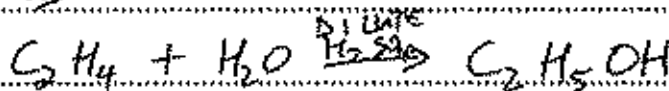
CATALYTIC CRACKING: LONGER CHAIN HYDROCARBONS are BROKEN into SMALLER MOLECULES



- (b) Ethanol can be formed from ethylene. Describe how this conversion is achieved and write an equation for the chemical reaction you describe.

2

ETHYLENE is HYDRATED in the presence of DILUTE SULFURIC ACID



Question 34 continues on page 25



## Question 34 (continued)

Marks

- (c) Ethylene can be distinguished from ethane by a simple laboratory test. Describe how this test is carried out and its results and include balanced equations for chemical reactions which occur in both the dark and in the light. Name the product formed.

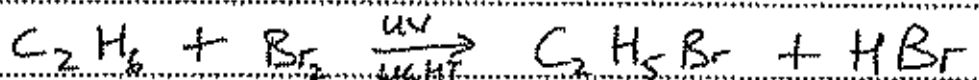
3

ETHYLENE  $C_2H_4$  is bubbled through some BROMINE water and the ORANGE/BROWN colour of BROMINE becomes COLOURLESS.  $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$

1,2-DIBROMOETHANE.

The above reaction occurs in both the dark + light.

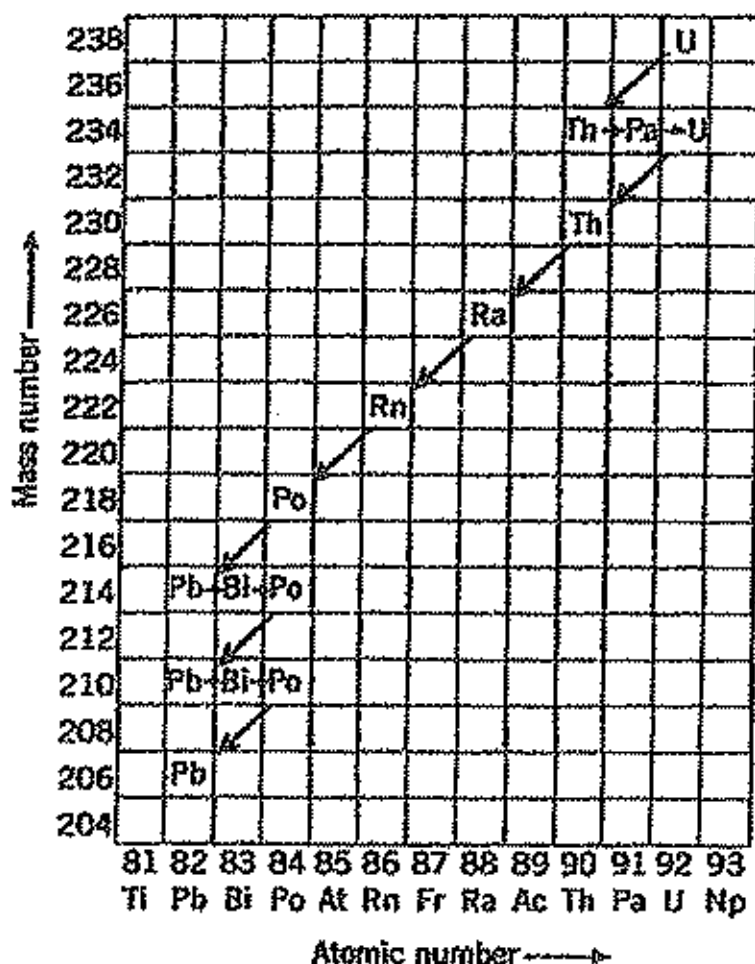
ETHANE  $C_2H_6$  is BUBBLED THROUGH some BROMINE water in the dark and no colour change is observed. However the ethane bubbled through bromine water in the PRESENCE OF LIGHT causes the BROMINE water to DECOLOURISE after a while.



## Question 35 (2 marks)

Marks

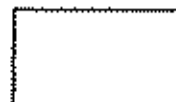
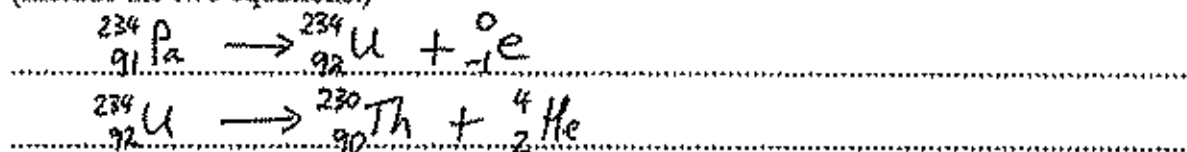
Uranium is a radioactive element that slowly undergoes decay.



Using the flow diagram shown above, trace the sequence of fission products during the decay of uranium for the steps Pa (Protactinium) to Th (Thorium) Th-230.

(Include the two equations.)

2



## Question 36 (3 marks)

Marks

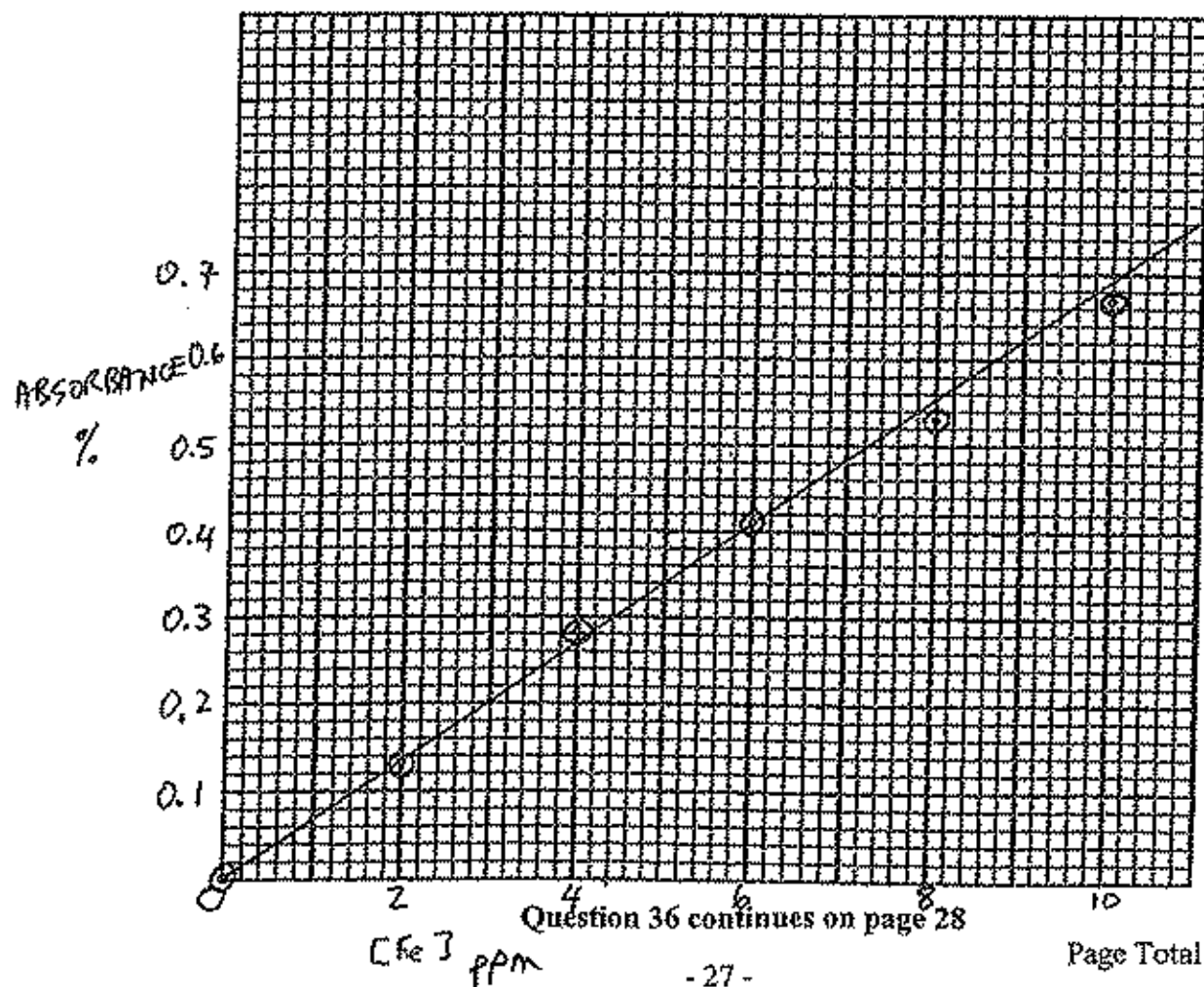
Railways often test the cooling water of their diesel engines for dissolved metals. The presence of particular metal ions in the water gives an indication as to the "health" of the engine. The most commonly used method for this analysis is atomic absorption spectroscopy.

In one particular analysis a chemist was testing for the presence of iron. She used a series of standard solutions to obtain the following results.

| Concentration of Fe (ppm) | Absorbance (%) |
|---------------------------|----------------|
| 0                         | 0.00           |
| 2                         | 0.13           |
| 4                         | 0.28           |
| 6                         | 0.41           |
| 8                         | 0.53           |
| 10                        | 0.67           |

- (a) Draw a calibration graph of these results.

2



## Question 36 (continued)

Marks

The chemist then tested the water from three engines.

For each engine she placed 50 mL of sample in a 1000 mL standard flask. The flask was then filled to the mark with distilled, deionised water and shaken to ensure mixing. She then tested each of the diluted samples and obtained the following results.

| Engine Number | Absorbance (%) |
|---------------|----------------|
| X12           | 0.04           |
| X45           | 0.01           |
| X67           | 0.30           |

- (b) Calculate the concentration of iron in the original sample from engine X67.

1

DILUTED SAMPLE 4.4 ppm is the concentration of Fe.

DILUTED 20X SO [Fe] IN ORIGINAL SAMPLE  $20 \times 4.4$   
 $= 88 \text{ ppm.}$

OR  $C_1 V_1 = C_2 V_2$

$$C_1 \times 50 = 4.4 \times 1000$$

$$C_1 = \frac{4.4 \times 1000}{50}$$

$$C_1 = 88 \text{ ppm.}$$

End of Paper

Page Total